DASIP - Difficulty Aware Stochastic Interpolant Policy

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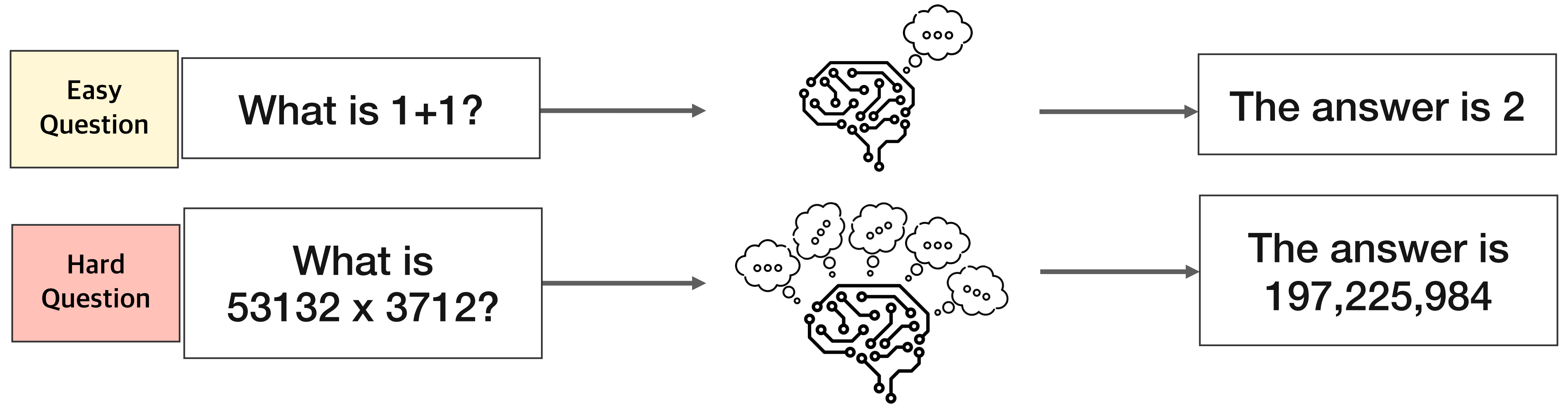
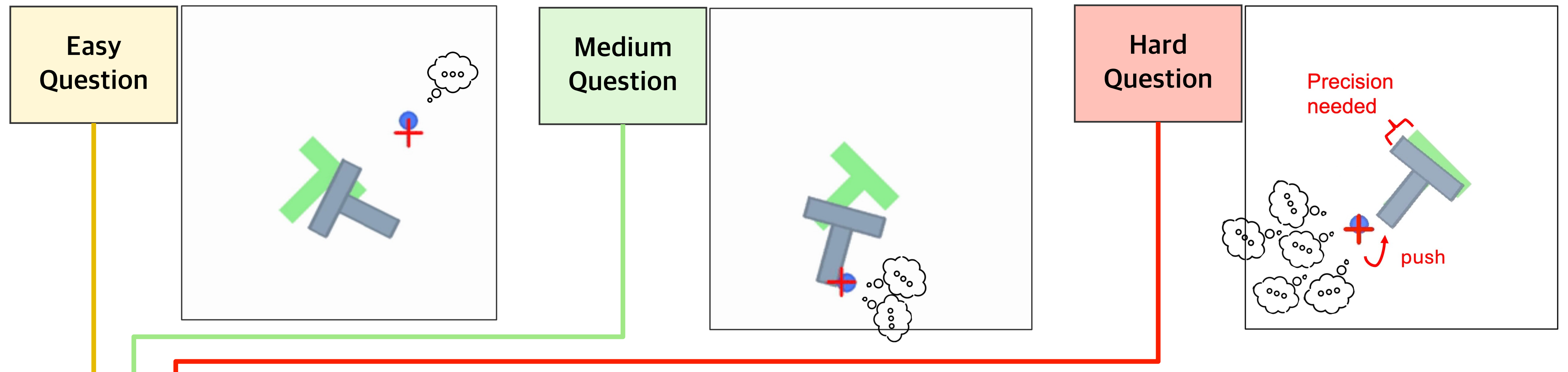
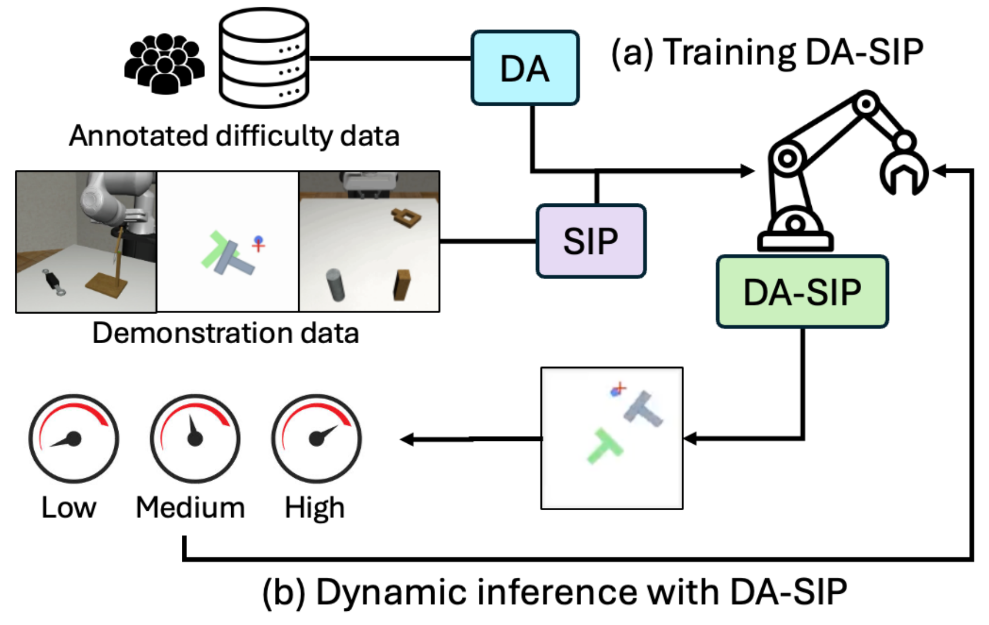
1. System 1 vs System 2 for robotics

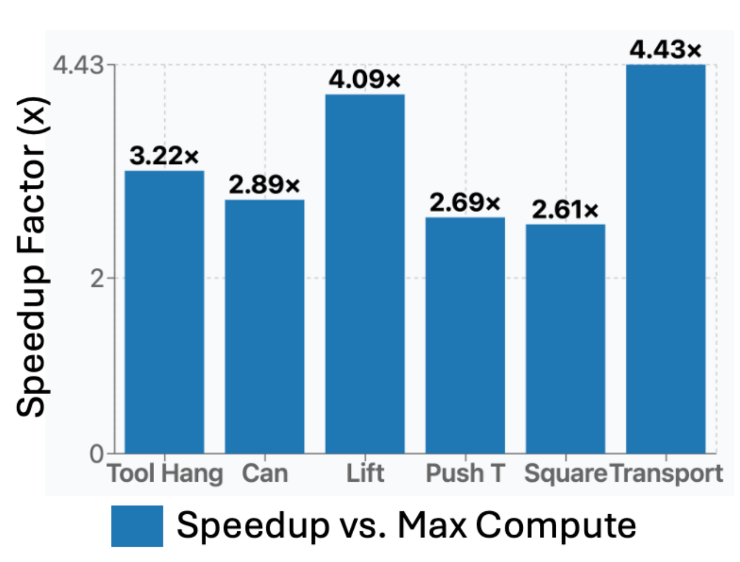

Easy Question
What is 1+1?
The answer is 2
Hard Question
What is 53132 x 3712?
The answer is 197,225,984


Easy Question
Medium Question
Hard Question
Precision needed
push

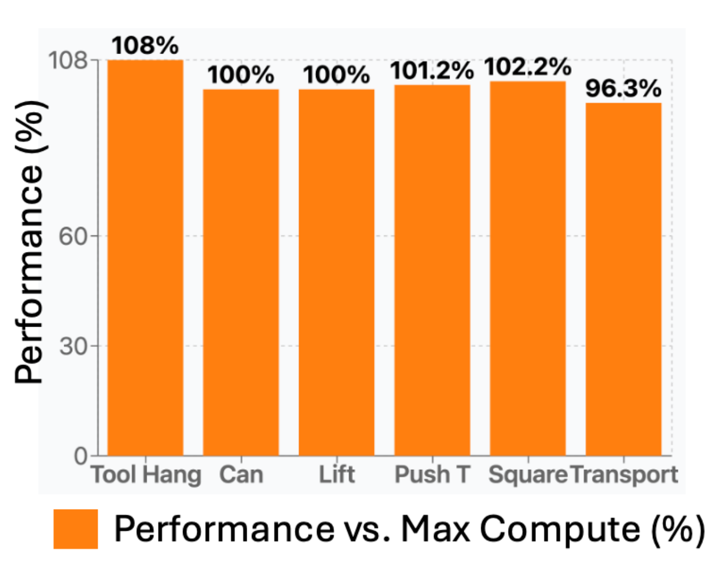
Category	Characteristics	Inference Steps	Solver	Integration
Initial state (I)	Robot positioned away from targets	**1**	**Euler**	**ODE**
Near (N)	Approaching within 30cm of targets	**5**	**Euler**	**ODE**
Grabbing (G)	Grasping and placing objects	**10**	**Euler**	**ODE**
Stochastic attempts (S)	Precise alignment requiring variability	**50**	**Euler**	**SDE**
Continuous pushing (C)	Sustained precise manipulation	**100**	**Heun**	**SDE**
End state (E)	Objectives achieved	**1**	**Euler**	**ODE**


DA
(a) Training DA-SIP
Annotated difficulty data
Demonstration data
SIP
DA-SIP
Low
Medium
High
(b) Dynamic inference with DA-SIP

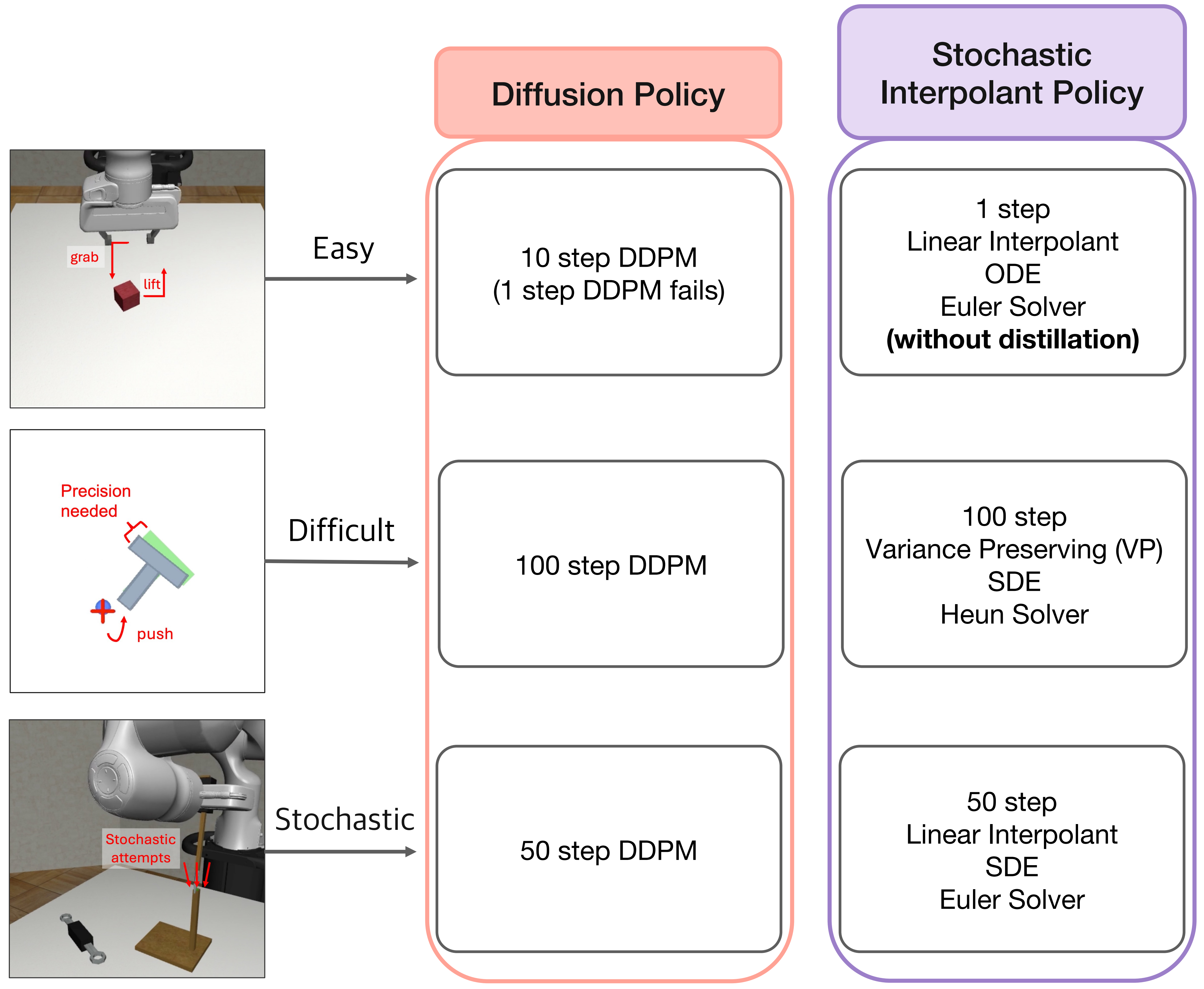
Compute Efficiency Gains



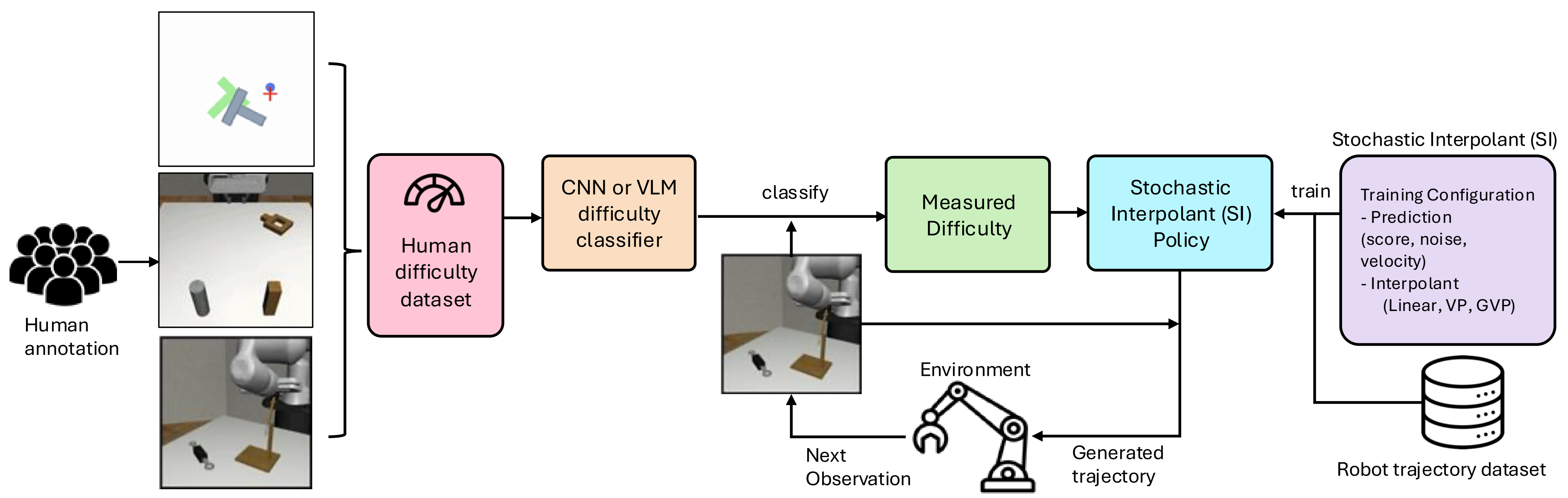
Performance Retention



2. SIP is a versatile unifying policy for robotics


grab
lift
Precision needed
push
Stochastic attempts

	Diffusion Policy	Stochastic Interpolant Policy
Easy	10 step DDPM (1 step DDPM fails)	1 step Linear Interpolant ODE Euler Solver **(without distillation)**
Difficult	100 step DDPM	100 step Variance Preserving (VP) SDE Heun Solver
Stochastic	50 step DDPM	50 step Linear Interpolant SDE Euler Solver


Human annotation
Human difficulty dataset
CNN or VLM difficulty classifier
classify
Measured Difficulty
Stochastic Interpolant (SI) Policy
train
Stochastic Interpolant (SI)
Training Configuration
- Prediction
(score, noise, velocity)
- Interpolant
(Linear, VP, GVP)
Robot trajectory dataset
Environment
Next Observation
Generated trajectory

3. Stochastic Interpolants (SI) unifies flow-based and diffusion-based methods

(such as Diffusion Model, Flow Matching, and Rectified Flow)

Stochastic Interpolant

Flow and diffusion based policies transform standard Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$ into data points $x_* \sim p(x \mid o)$, where o are robot observations. These approaches can be unified through the interpolation formulation.

$$x_t = \alpha_t x_* + \sigma_t \varepsilon, \tag{1}$$

where α_t is increasing and σ_t is decreasing, interpolating from noise to data regimes.

Velocity. The conditional velocity of the bridge is

$$v(x, t, o) = \mathbb{E}[\dot{x}_t \mid x_t = x, o] = \dot{\alpha}_t \mathbb{E}[x_* \mid x_t, o] + \dot{\sigma}_t \mathbb{E}[\varepsilon \mid x_t, o]. \tag{2}$$

Score. Because Eq. (1) is linear–Gaussian, Bayes' rule gives a one-line posterior:

$$\mathbb{E}[\varepsilon \mid x_t = x, o] = \frac{\alpha_t v(x, t, o) - \dot{\alpha}_t x}{\dot{\alpha}_t \sigma_t - \alpha_t \dot{\sigma}_t}.$$

Substituting yields

$$s(x, t, o) = \sigma_t^{-1} \frac{\alpha_t v(x, t, o) - \dot{\alpha}_t x}{\dot{\alpha}_t \sigma_t - \alpha_t \dot{\sigma}_t}. \tag{3}$$

Reverse-time SDE. Using the velocity and score fields learned within the stochastic interpolant framework, we apply the reverse-time SDE theory extended to the conditional density $p_t(x \mid o)$:

$$dX_t = \left[v(X_t, t, o) - \tfrac{1}{2} w_t s(X_t, t, o)\right] dt \; + \; \sqrt{w_t}\, d\bar{W}_t. \tag{4}$$

Interpolation Methods

Linear	Variance-Preserving (VP)	generalized VP (GVP)
$\alpha_t = t, \qquad \sigma_t = 1 - t.$ A straight path suited to well-behaved trajectories.	$\alpha_t = \sqrt{1 - \exp\left(-\int_0^t \beta_s \, \mathrm{d}s\right)},$ $\sigma_t = \exp\left(-\frac{1}{2} \int_0^t \beta_s \, \mathrm{d}s\right).$ A noise schedule β_t adds Gaussian noise while preserving total variance.	$\alpha_t = \sin\left(\frac{\pi}{2} t\right), \; \sigma_t = \cos\left(\frac{\pi}{2} t\right).$ Trigonometric path with $\alpha_t^2 + \sigma_t^2 = 1$ for all t, yielding smoother gradients and better stability.